

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is the highest-priority dimension per elicitation (HIGH), and dataset analysis reveals a critical concern: all 26 sampled Hindi validation examples are flagged is_translated=True, suggesting the Hindi validation split is dominated by GPT-4O-translated English-source items rather than natively-sourced Hindi competitive-exam questions. While MILU's documentation claims an India-first design **[Q11]** and includes regionally relevant subjects **[Q12, Q14]**, the actual Hindi content sampled shows: (a) literary content focused on Wordsworth rather than Tulsidas/Premchand/Kabir **[DATASET-D5]**; (b) translated technical terminology using transliterations (शेल, नाइट्रेट, जंप लेबल) that may not align with state-board Hindi textbook conventions **[DATASET-D14, D15, D23]**; (c) only one item with explicit North Indian state-specific content (UP minerals) **[DATASET-D2]**; (d) parallel translated items appearing across all 11 language configs **[DATASET-D11, D18, D19, D24]**. The paper's dataset-level 25% translation figure **[Q47]** underestimates the apparent Hindi-validation-split translation rate.

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q14	"We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D5	विलियम वर्ड्सवर्थ _____ के कवि हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature
D11	What was the first capital of the Bahamani Kingdom? ... Gulbarga — Medieval Indian history — source question in English
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिनलखिति में से किसकी अधिकता 'ब्लू बेबी सिंड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check
D23	फॉरट्रान 77 के फकिस्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary
D24	एक भौगोलिक क्षेत्र में नमिनलखिति वशिष्ट वशिष्टाएँ हैं: ... उत्तरी अमेरिका का अटलांटिक तट — Mediterranean climate geography — translated, no North India content

Strengths

- Documented India-first design intent with stated coverage of local history, arts, festivals, and laws **[Q2, Q11, Q12]**
- Region-specific exams are explicitly emphasized for capturing local knowledge **[Q14, Q28]**

- Some India-centric Hindi items present (UP minerals, Mughal history, Amrita Shergil, Radcliffe Line) [DATASET-D2, D3, D8, D25]
- Cross-language answer-key consistency demonstrated for translated parallel items [DATASET-D18, D19]

ID	Evidence
Q2	“With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science.” (p.1)
Q11	“We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.” (p.2)
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
Q14	“We focus on region-specific exams to authentically capture local knowledge in the respective language.” (p.2)
Q28	“Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state.” (p.3)
D2	नमिन्लखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
D3	मुगल ग्रंथ 'मासि-ए-आलमगीरी' के रचयिता कौन हैं? साकी मुसतईद खान — Mughal history — competitive-exam GK register
D8	नमिन्लखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार हैं, जिन्हें कभी-कभी भारत की फ्रीडा काहलो के नाम से जाना जाता है? अमृता शेरगिल — Indian arts & culture — Amrita Shergil, India-centric
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D19	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check
D25	रेडक्लिफ रेखा भारत की सीमाओं को नमिन्लखिति में से किस देश के साथ विभाजित करती है? पाकिस्तान — Partition history — India-centric GK

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires region-specific knowledge: North Indian state geography, Hindi-board canonical literature, Panchayati Raj civics, state freedom-movement figures. The Hindi sample contains some India-centric items [DATASET-D2, D3, D25] but a majority of items are translated [DATASET-D28] and may not reflect native Hindi competitive-exam content.

IC-2: Cultural alignment is partially present (Mughal history, Indian arts, partition geography are reasonable matches [DATASET-D3, D8, D25]), but the Hindi-literature register [DATASET-D5 — Wordsworth as the only literature item] does not match Hindi-medium school-board norms. Canonical Hindi literary figures (Tulsidas, Premchand, Kabir, Mahadevi Varma) are not present in the sampled items.

IC-3: Translated items retain English technical transliterations (शेल, नाइट्रेट, प्लूसेटिंग डीसी, जंप लेबल) [DATASET-D14, D15, D23] that may diverge from Vaigyaanik evam Takniki Shabdavali Aayog or state-board conventions. The Wordsworth literature item [DATASET-D5] introduces non-regional content. Translated items effectively test translated-English knowledge rather than native Hindi-register knowledge.

IC-4: INSUFFICIENT DOCUMENTATION — MILU does not document recruitment of Hindi-medium North Indian annotators specifically [NOT FOUND per regional YAML annotator_demographics]; institutional base is IIT Madras + IBM Research India, which does not guarantee North Indian Hindi-medium representation.

IC-5: Documented content issues: (a) ~100% translation rate observed in Hindi validation sample vs. ~25% claimed dataset-wide [Q47, DATASET-D28]; (b) absence of canonical Hindi-board literary content; (c) under-representation of UP/Bihar/Rajasthan/MP state-specific civics; (d) translation terminology not validated against state-board Hindi norms [NOT FOUND per regional YAML hindi_register_standardisation].

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Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
Q84	"Third, the scarcity of questions necessitated translating a portion of the dataset." (p.9)
D2	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question
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D28	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... is_translated: True — is_translated=True flag on all 26 Hindi validation examples reviewed
WEB-19	Bihar's majority population speaks Bhojpuri (24.86%), Maithili (12.55%), Magahi (10.87%) — Khari Boli register may amplify mismatch for this sub-population (https://en.wikipedia.org/wiki/Bihar)

Information Gaps

- Whether the test split is also predominantly is_translated=True (only validation split sampled)
- Hindi-specific translation proportion in MILU (paper reports only dataset-wide 25%)
- State-wise distribution of Hindi exam sources (UP/MP/Rajasthan/Bihar shares)
- Whether Hindi literature items in the un-sampled portion include canonical board authors

Requires Expert Verification

- Whether GPT-4O Hindi translations preserve subject-specific terminology in line with UP/MP/Rajasthan/Bihar state-board textbooks
- Whether Khari Boli register mismatches with Bhojpuri/Maithili-speaking Bihar teachers materially affect comprehension
- Whether 'शेल', 'नाइट्रेट', 'जंप लेबल' style transliterations are acceptable in state-board Hindi technical texts

Remediation

Gap 1: 100% of sampled Hindi validation items are machine-translated; translated technical terminology uses English transliterations of unvalidated alignment with state-board Hindi norms; canonical Hindi-board literary content absent from sample

Recommendation: Stratify any deployment-relevant evaluation by is_translated flag and report performance separately for natively-sourced vs. translated Hindi items; supplement with a curated set of native Hindi-board MCQs covering Hindi

literature canonical authors and verified state-board terminology

Gap 2: No published state-wise breakdown of Hindi exam sources for UP/MP/Rajasthan/Bihar

Recommendation: Request access to the gated HuggingFace dataset metadata (or supplementary Table 8) and audit the Hindi item provenance by source-exam state; if Hindi-belt state-PSC sourcing is thin, supplement with state-PSC items directly